

SpellCam

GitHub: github.com/KimNattanan/spellcam | Website: 13-126-39-3.sslip.io

A **video call web application** where users sign in, create custom hand gestures, and on a peer-to-peer video call, posing created hand gestures will displays the label for both people. Hand poses are detected with MediaPipe Hand Landmarker from each caller's camera feed. By setting the gesture label to "[[anime]]", a random anime image URL fetched from <https://nekosia.cat/> will be displayed instead of the label.

FEATURES

- **Authentication** – Register and sign in with credentials stored in MongoDB. Sessions use signed JWTs delivered via HTTP-only cookies.
- **Custom gesture recording** – Lets each user create, update, delete, and record their own gesture templates (1-hand, 2-hand, and 2-hand-relate) stored in MongoDB.
- **Peer video calls** – WebRTC (camera + microphone) with Socket.IO signaling on the same process as the Next.js app. Canonical rooms pair two authenticated users.
- **Gesture sync** – Over a WebRTC data channel, each peer sends the latest predicted label and, only for "[[anime]]", the fetched anime image URL so the remote feed can render the same picture, and the partner relies on the synced label for matching text.

TECH STACK

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| • Languages: | TypeScript |
| • Libraries / Frameworks: | Next.js, TailwindCSS, Shadcn (Radix UI), Node.js, Socket.IO, MediaPipe (Hand Landmarker) |
| • APIs: | RESTful, WebRTC |
| • Databases: | MongoDB |

OPTIONAL TIER CHOSEN

- S – Computer Vision

TEAM MEMBERS

- Nattanan Jeungtanasirikul
- Kanasit Bunkun
- Sutiwat Dachdachasunun